

# Prefix Tuning

*Trainable soft prompts prepended to every Transformer layer's key-value space — no weight modification*

**9 min**

Duration

**B3 – Apply**

Bloom's

**L3 – Advanced**

Level

**AI-FT-IN-TH-000008**

ID

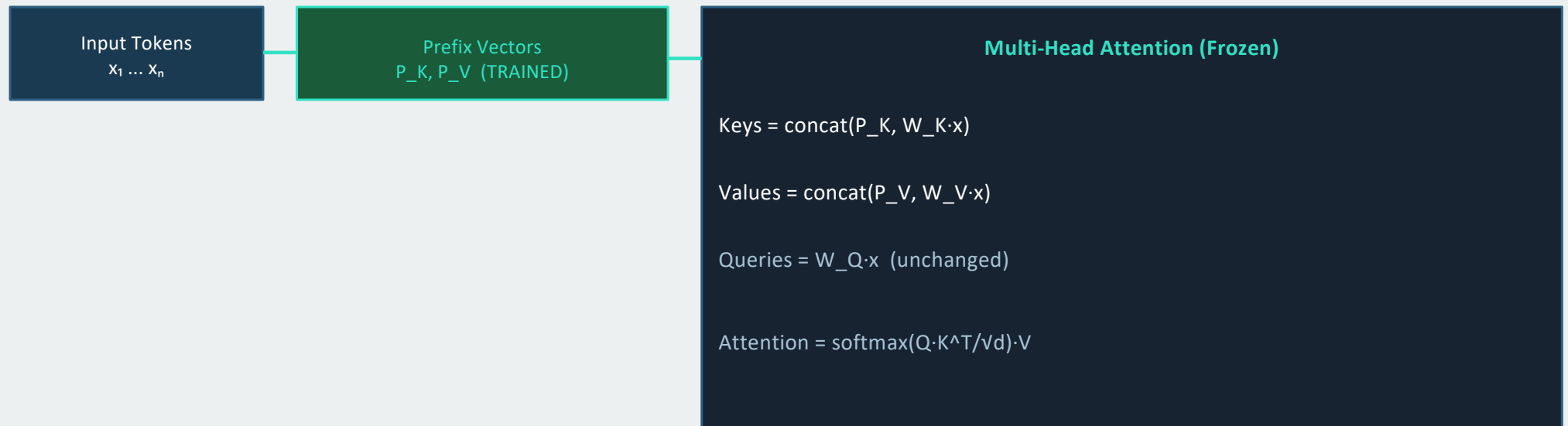
## Prefix Tuning — Core Idea

### Definition (Li & Liang, ACL 2021)

Prefix Tuning prepends a sequence of trainable continuous vectors (the 'prefix') to the key-value pairs at every Transformer layer. Unlike hard prompts (discrete tokens), prefix vectors are continuous and optimised directly via gradient descent.

- The prefix is not constrained to the discrete token vocabulary — it can represent any direction in activation space
- Parameterised through a reparameterisation MLP during training for stability:  $\text{prefix} = \text{MLP}(\text{prefix\_params})$ . The MLP is discarded after training — only the prefix embeddings are stored
- Number of trainable parameters:  $L \times P \times 2 \times d$  ( $L$  = layers,  $P$  = prefix length,  $d$  = model dimension)
- Fully separable from the base model: the base weights are never touched

## Prefix Tuning — Architecture Diagram



This same prefix is applied at every Transformer layer —  $P_K$  and  $P_V$  are layer-specific. The prefix acts as a soft, task-specific 'memory' that conditions the model's attention at every layer simultaneously. This is more powerful than a single soft prompt at the input only (prompt tuning).

# Prefix Tuning vs Hard Prompts vs Soft Prompt Tuning

## ✓ Advantages over Hard Prompts

- Continuous optimisation — not constrained to real tokens
- Can represent any concept, not just expressible vocabulary
- End-to-end optimised for the task — no manual prompt engineering
- Gradient signal is informative — no straight-through estimator needed

## ⚠ Disadvantages vs LoRA

- Consumes context window:  $P$  prefix tokens used per call at inference
- Generally underperforms LoRA on tasks requiring deep parameter adaptation
- Training can be unstable without the reparameterisation MLP trick
- Prefix length is a hyperparameter requiring tuning (typically 10–100 tokens)

## Prefix Tuning Hyperparameters

| Parameter                         | Typical Range                                      | Effect                                                               |
|-----------------------------------|----------------------------------------------------|----------------------------------------------------------------------|
| Prefix length P                   | 10–100 tokens                                      | Longer prefix = more capacity but more context consumed at inference |
| Reparameterisation MLP hidden dim | 512–2048                                           | Larger MLP = more stable training; discarded post-training           |
| Prefix dropout                    | 0.0–0.1                                            | Regularisation; prevents overfitting to training distribution        |
| Applied layers                    | All transformer layers (standard)                  | Applying to all layers is key; input-only prefix is much weaker      |
| Task format                       | Constrained generation (table→text, summarisation) | Best performance on structured output tasks                          |

## KEY REFERENCES

- Li, X.L. & Liang, P. (2021). Prefix-Tuning: Optimizing Continuous Prompts for Generation. ACL 2021.
- Lester, B. et al. (2021). The Power of Scale for Parameter-Efficient Prompt Tuning. EMNLP 2021.
- Liu, X. et al. (2022). P-Tuning v2: Prompt Tuning Can Be Comparable to Fine-Tuning. arXiv.

 **Next: Concept 10: Adapter Modules (AI-FT-IN-TH-000009)**